

MVLU COLLEGE

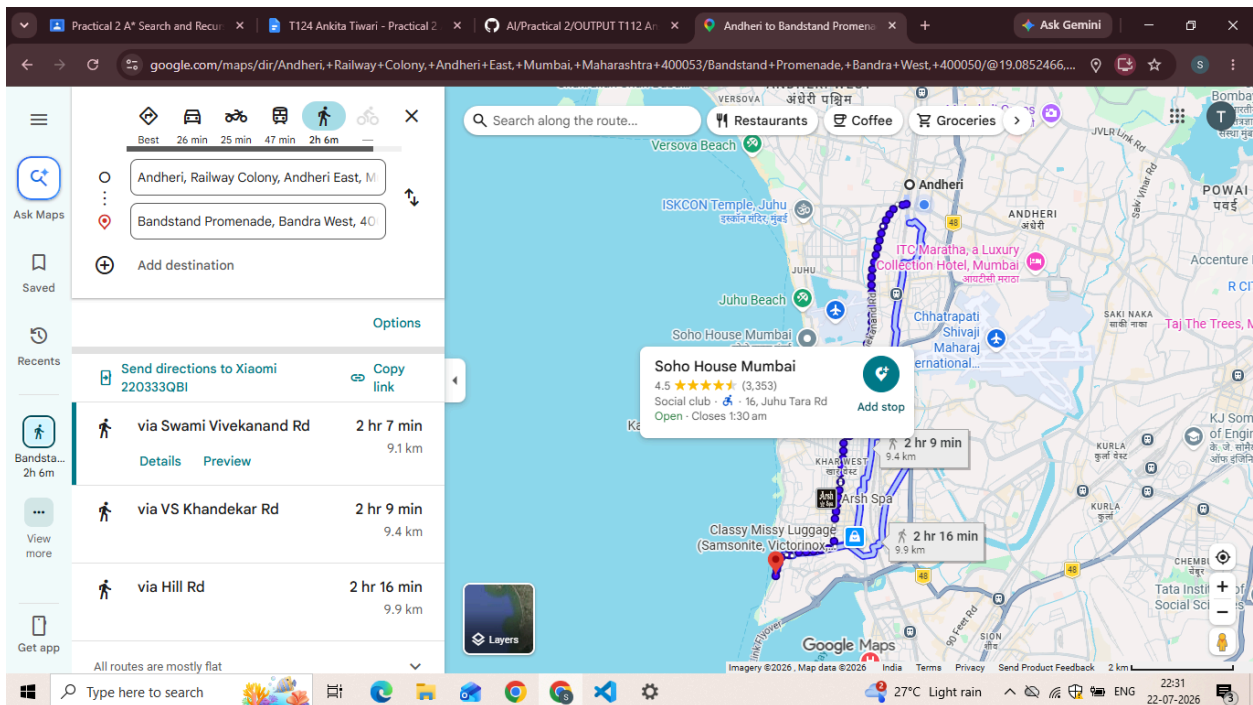
AIM:

- Implement the A* Search algorithm for solving a pathfinding problem.
- Implement the Recursive Best-First Search algorithm for the same problem.
- Compare the performance and effectiveness of both algorithms.

OUTPUT:

Recursive Best-First Search:

MAP:



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T124

TYCS

AI PRACTICAL NO.2

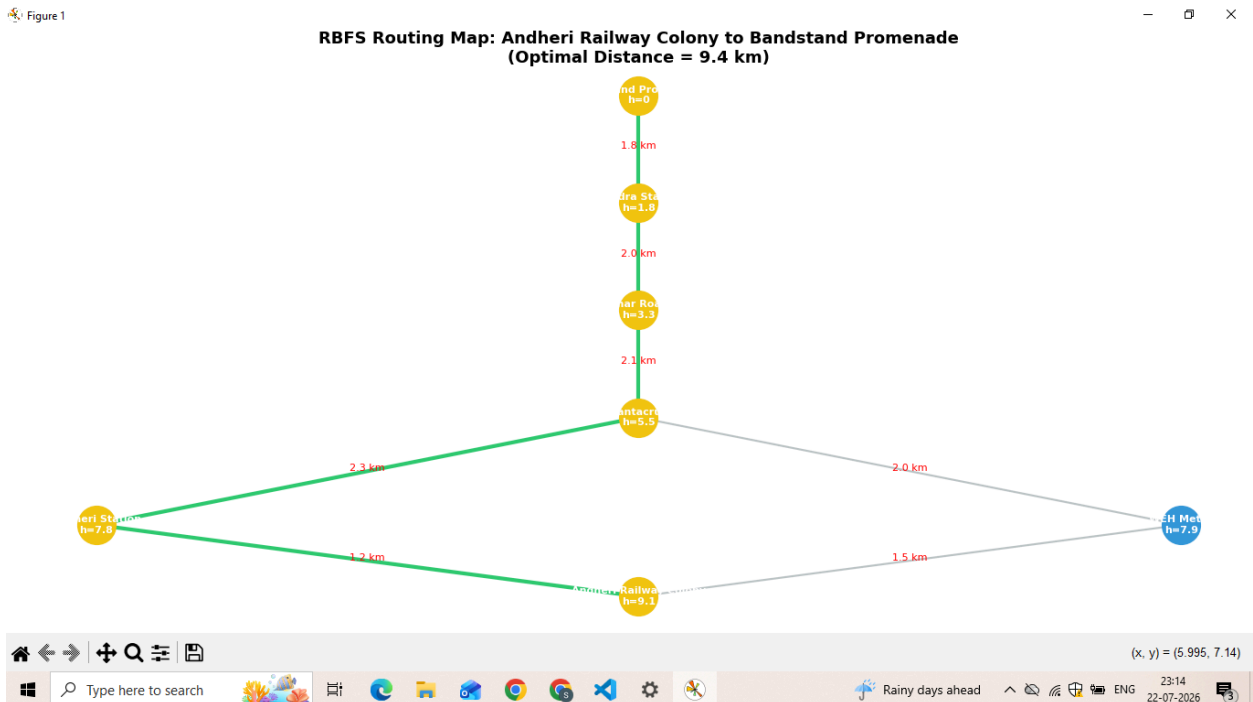
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```
File Edit Selection View Go Run ... Search Update
practical_2_breadth first search.py X
C:\Users\Admin\Desktop> AI > practical_2_breadth first search.py > ...
236     fontsize=8,
237     fontweight='bold'
238 )

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Python 3.13 (64-bit)
PS C:\Users\Admin\Desktop\Microsoft VS Code> & C:\Users\Admin\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Admin/Desktop/AI/practical_2_breadth first search.py"
Optimal RBFS Path:
Andheri Railway Colony -> Andheri Station -> Santacruz -> Khar Road -> Bandra Station -> Bandstand Promenade

Total Road Distance: 9.4 km
Ln 55, Col 28 Spaces: 4 UTF-8 CRLF Python Python 3.13 (64-bit)
Type here to search 4 cm of rain Thu 23:17 22-07-2026
```

GRAPH:



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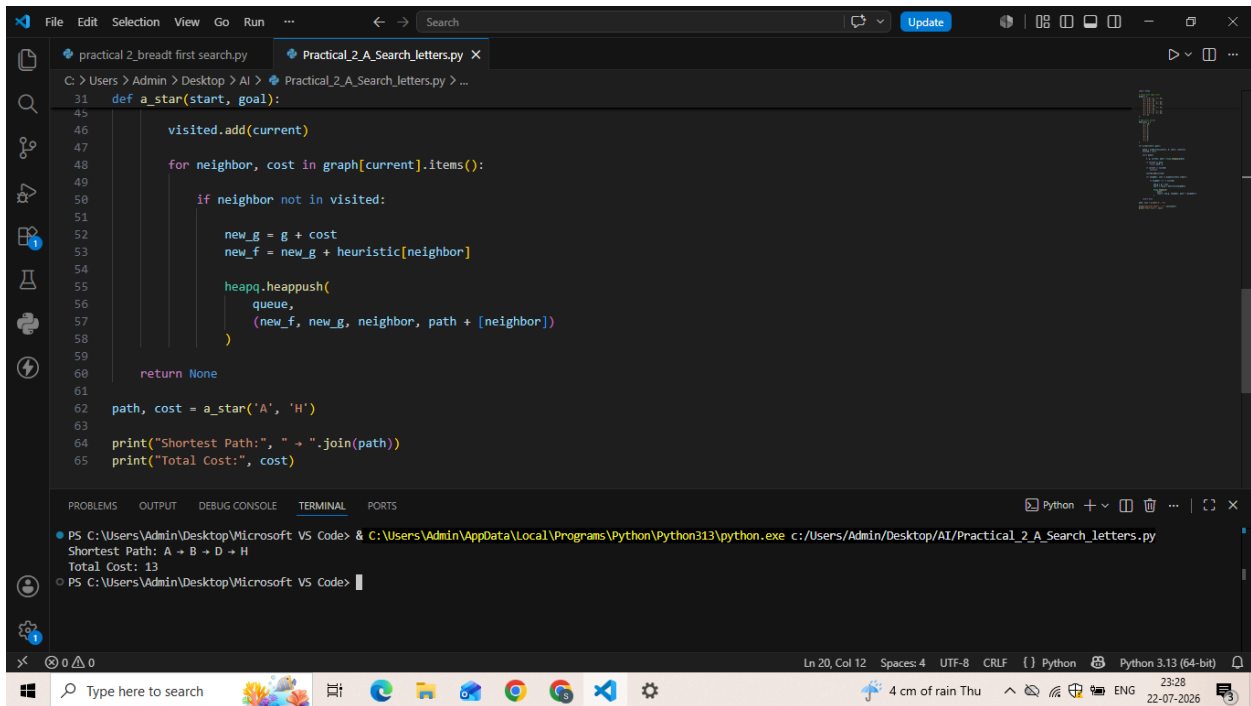
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A* SEARCH (LETTERS):



```
def a_star(start, goal):
    visited.add(current)
    for neighbor, cost in graph[current].items():
        if neighbor not in visited:
            new_g = g + cost
            new_f = new_g + heuristic[neighbor]
            heapq.heappush(queue, (new_f, new_g, neighbor, path + [neighbor]))
    return None

path, cost = a_star('A', 'H')
print("Shortest Path:", " -> ".join(path))
print("Total Cost:", cost)
```

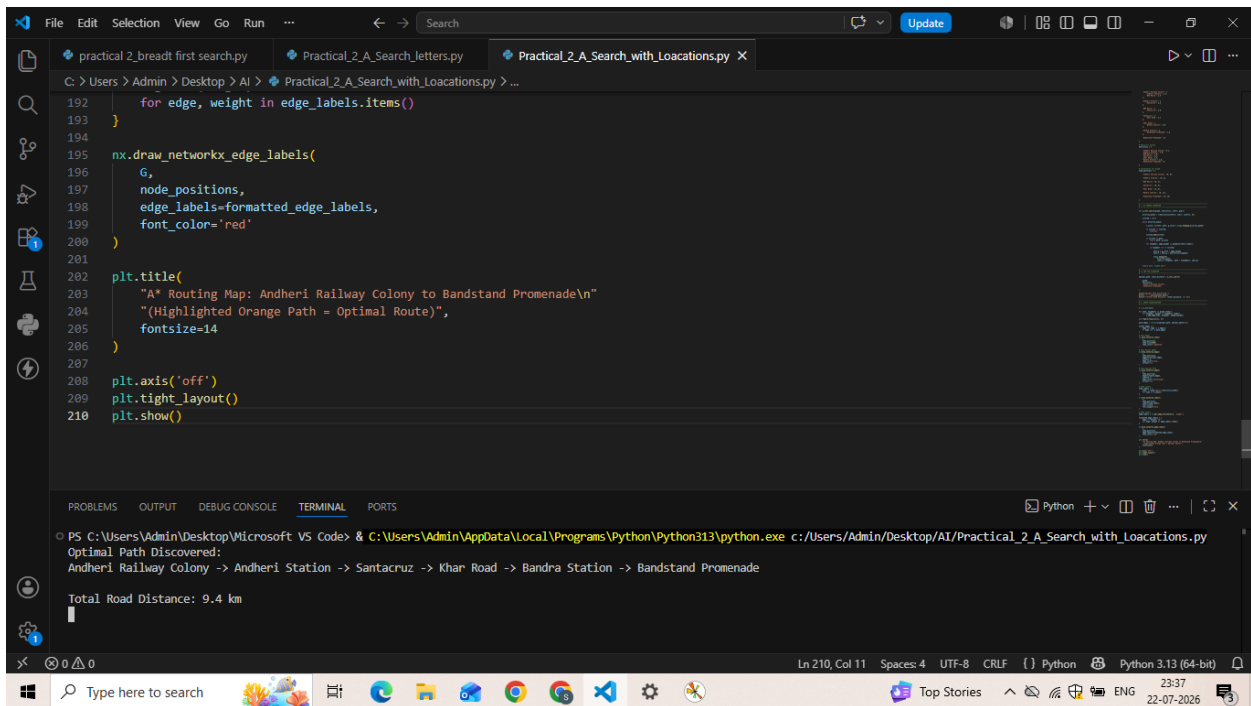
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Admin\Desktop\Microsoft VS Code> & C:\Users\Admin\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Admin/Desktop/AI/Practical_2_A_Search_letters.py

Shortest Path: A -> B -> D -> H
Total Cost: 13

PS C:\Users\Admin\Desktop\Microsoft VS Code> |

A* SEARCH (LOCATION):



```
for edge, weight in edge_labels.items()
}
mx.draw_networkx_edge_labels(
    G,
    node_positions,
    edge_labels=formatted_edge_labels,
    font_color='red'
)
plt.title(
    "A* Routing Map: Andheri Railway Colony to Bandstand Promenade\n"
    "(Highlighted Orange Path = Optimal Route)",
    fontsize=14
)
plt.axis('off')
plt.tight_layout()
plt.show()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Admin\Desktop\Microsoft VS Code> & C:\Users\Admin\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Admin/Desktop/AI/Practical_2_A_Search_with_locations.py

Optimal Path Discovered:
Andheri Railway Colony -> Andheri Station -> Santacruz -> Khar Road -> Bandra Station -> Bandstand Promenade

Total Road Distance: 9.4 km

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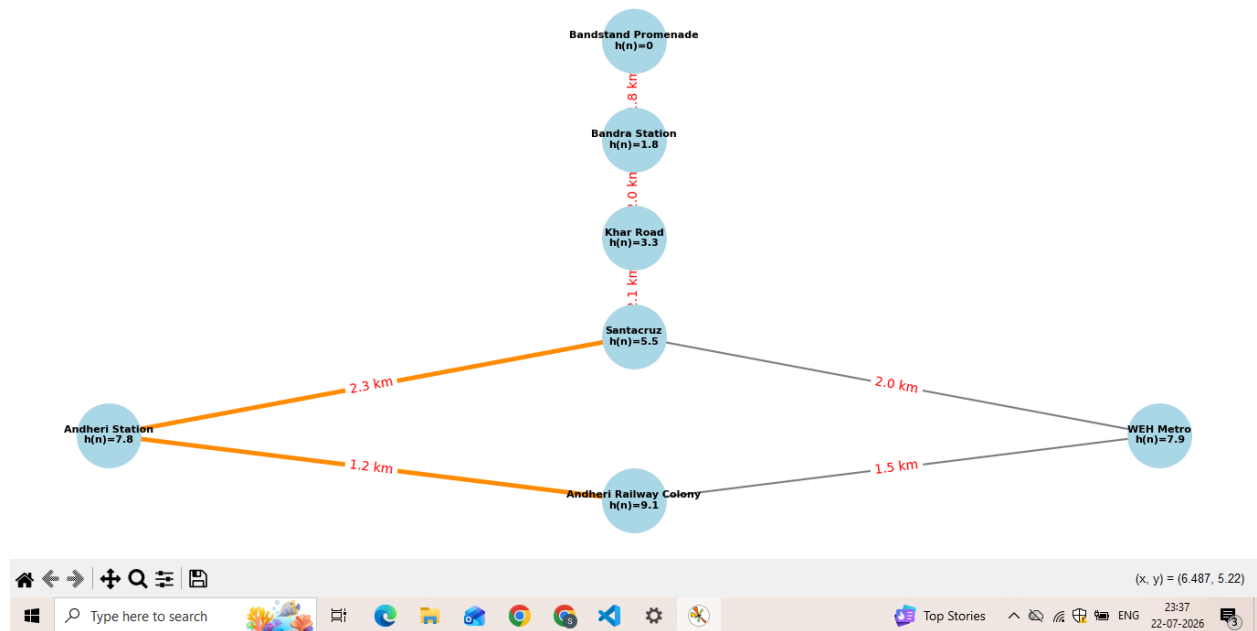
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GRAPH:

Figure 1

A* Routing Map: Andheri Railway Colony to Bandstand Promenade
(Highlighted Orange Path = Optimal Route)



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